

Fundamentals of Deep Learning and Programming

Lecture 0: Course Overview

Fall 2025

Adjunct Professor: Donghoon Kang

Course Overview

Lecture Time

Fridays, 2 PM – 5 PM

Classroom

Room: L3 301, 302 (3334A, B)

Evaluation Criteria

Attendance (10%)

Programming Assignments (15%)

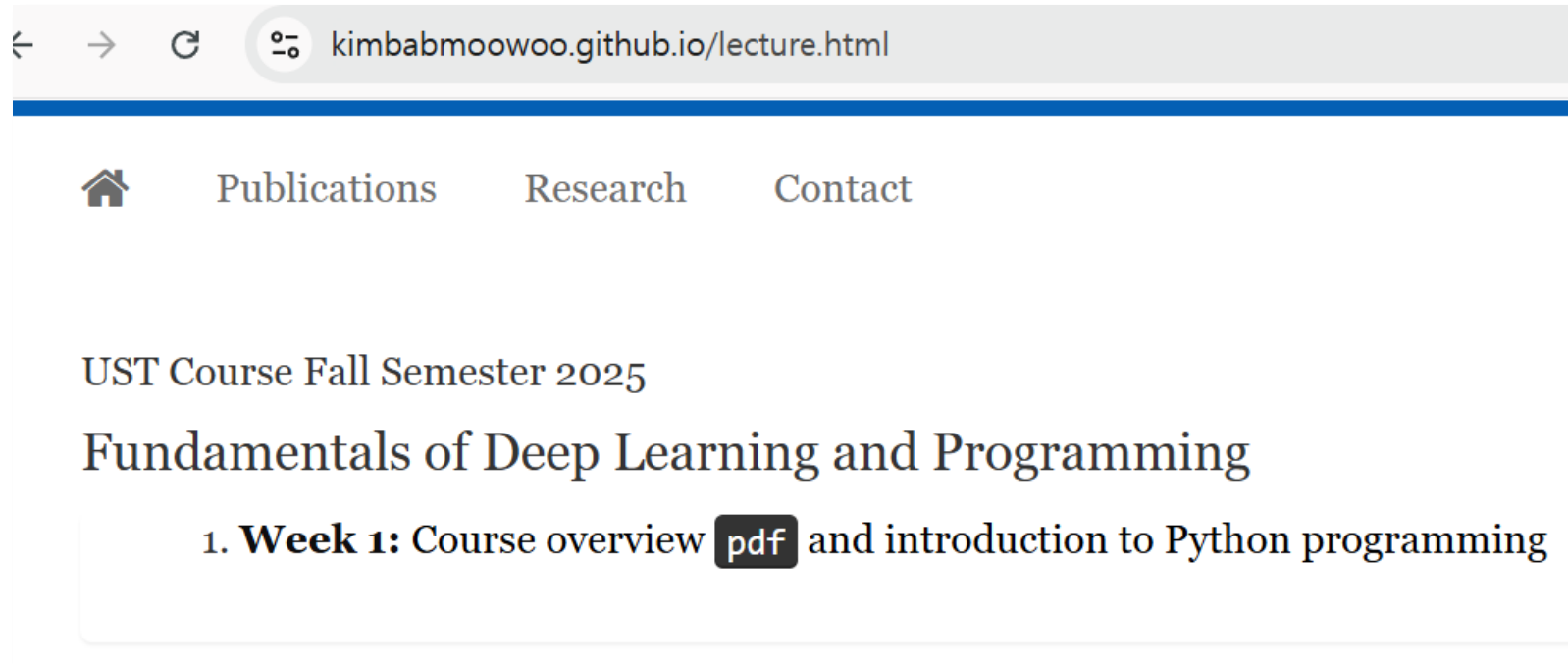
Term-project (15%)

Mid-term (30%), Final (30%)

- Programming assignments: students should submit the **code** and a **2-3 page document** briefly explaining the code to me by **email** (chocopie@kist.re.kr)

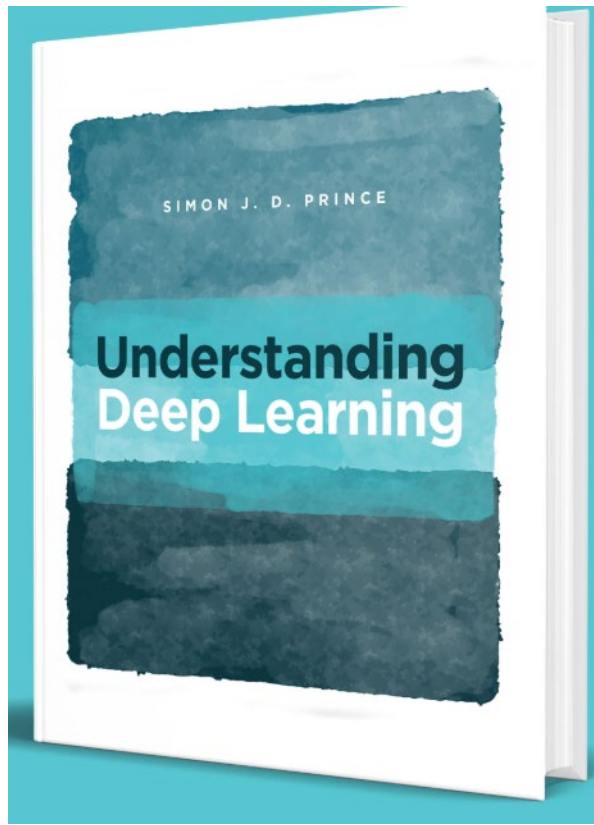
Course Website

<https://kimbabmoowoo.github.io/lecture.html>



Recommended book (pdf is free)

<https://udlbook.github.io/udlbook/>



Understanding Deep Learning

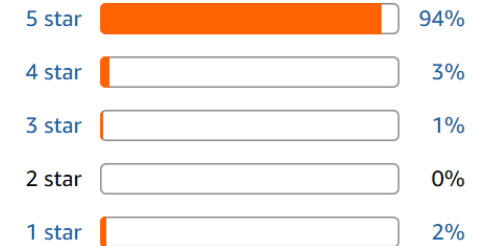
by Simon J.D. Prince | Dec 5, 2023

★★★★★ 168

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★★★★★ 4.9 out of 5

168 global ratings



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Term Project

Must be performed **alone**

- Should **select a paper** published only in **CVPR, ICCV, ECCV, or NeurIPS after 2023 (inclusive)**.
- Should **write an 8-10 page report** analyzing the training code of the selected paper.
(including the **experimental results** using **a subset** of training data)

Before Week 7

Should decide which **paper** to **choose**. (also possible to change another paper later)

Before Week 15

Should **submit** an 8-10 page **report** as a term-project.

Schedule

September

Week 1

Course overview

Introduction to Python Programming

Week 2

Useful Python Libraries

Numpy, PyTorch

Week 3

Neural Networks Basics

Week 4

Convolutional Neural Networks (CNNs)

Schedule

October

Week 5

Models using CNN

Week 6

Introduction to Natural Language Processing (NLP)

Week 7

Attention Mechanism

Week 8

Transformer

Part 1

Week 9

Mid term exam

Schedule

November

Week 10

Transformer

Part 2

Week 11

Transformer

Part 3

Vision Transformer (ViT)

Week 12

Generative Models – Introduction to GANs, VAEs

Week 13

Diffusion Models

Part 1

Schedule

December

Week 14

Diffusion Models

Part 2

Week 15

Diffusion Models

Part 3

Week 16

Final exam